

MathSearch

Search a PDF for mathematical symbols by how they look

Guide and test sheet

Version 1.0

This document opens when the MathSearch viewer starts. It explains how to use the extension and is also a practice sheet: the formulas in it exist only so that there is something to search for. They are made up and state nothing of mathematical interest.

To search your own PDF, open it with the viewer's **Open** button (in the **Tools** menu at the top right) or press `Ctrl+O`, or drag the file onto the viewer. To read several PDFs at once, click the MathSearch toolbar icon again: each click opens a new viewer tab with its own search.

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1 Getting started

The MathSearch panel floats over the page, at the top right when the viewer opens. Drag it by its header to move it out of the way, and double-click the header to put it back. The ▼ button collapses it and × hides it. The Σ button in the viewer's toolbar, or **Alt+M**, brings it back.

1. Type the symbol as LaTeX in the box, for example `\mathfrak{F}`. The preview underneath shows it as it will be searched for: what you see is the search template.
2. Press **Enter** or click **Search**. Every place in the document where the symbol *looks* the same is highlighted in yellow, and the current match in orange, each with its score.
3. Press **Enter** again to go to the next match and **Shift+Enter** to go back, or click an entry in the list.

Searching by snip

If you don't know a symbol's LaTeX, click the scissors button next to the box (or press **Alt+S**) and drag a rectangle around the symbol on the page. The snip replaces the preview and is searched for straight away. It stays the query until you type in the box again. Snip only the symbol: anything else inside the rectangle, such as a subscript, a fraction bar or the edge of a neighbouring letter, is searched for too and lowers the scores of clean matches. **Esc** cancels a snip.

Keyboard shortcuts

Alt+M	show the panel and put the cursor in the search box
Enter	search; once there are results, next match
Shift+Enter	previous match
Ctrl+Enter	look the symbol up in the book's index first
Alt+S	snip a symbol from the page and search for it
Esc	leave the search box
double-click header	move the panel back to the top right

Settings in the panel

Scope searches the whole document or only the current page. **Min score** (0.90 by default) hides weaker matches. Lowering it after a search shows more candidates at once, without searching again. This helps with small occurrences, such as a symbol inside a subscript. **Penalise scripts and accents next to the match** (on by default) makes a symbol that carries an extra sub- or superscript, accent or dot rank below the bare symbol (see Section 2). Untick it to find the symbol with or without them; the search runs again, because the scores change. The setting lasts until you close the viewer.

Memory settings

Open **Settings** at the bottom of the panel. These two are kept in your browser between sessions; nothing else is stored. **Memory for the page index** (250 MB by default, at least 50) limits how much memory the extension keeps for pages it has already prepared for searching. A page of text takes about 0.16 MB, so most books fit. In a longer book, the extension keeps the book's index and list of notation and the pages around where your last search started, half before and half after;

other pages are prepared again when a search reaches them, which is slower but finds the same matches. The line under the setting shows how much is in use. **Release it when the tab is hidden for** (30 minutes by default; also 10 minutes, 1 or 2 hours, or never) gives the memory back while you work in another tab. It is prepared again when you come back.

2 How the search works

MathSearch does not read the text of the PDF: it compares shapes. Each page is rendered as an image, and the symbol from the preview is looked for in it, at any size. This finds symbols that have no text form in the PDF, and it tells apart letters that text search treats as the same, such as $\mathfrak{F}$, $\mathcal{F}$, $\mathcal{F}$ and F .

Search order. The search starts at the page you are reading and goes backward to the first page, then forward from where you started to the end. A symbol is usually defined before it is used, so the nearest earlier occurrence, often its definition, is found first. The viewer jumps to the first match as soon as it is found, and the rest appear while the search continues. **Cancel** stops it and keeps what was found.

How long it takes. After a book is opened, its pages are prepared in the background (“Indexing pages 3/24 . . .”); a search does not wait for that, but goes faster once it is done. A search through a whole prepared book takes a few hundredths of a second per page, longer for larger or compound symbols: in a 1100-page textbook on a laptop, about 40 s for $\mathcal{X}$ and about 90 s for a snip of $L(H)$. Pages beyond the memory limit (see *Memory settings*) are prepared again during the search, which adds to that: the same $\mathcal{X}$ search took about 70 s with the limit set to 50 MB. Because the search starts at your page, the nearest occurrence usually shows up within seconds, long before the search ends.

Sub- and superscripts count. Searching for $\mathbb{R}$ ranks $\mathbb{R}^d$ and $\hat{R}$ below a bare $\mathbb{R}$, because the extra marks on the page are not part of what you asked for. To find $\mathbb{R}^d$ itself, type `\mathbb{R}^d`; to find $\mathbb{R}$ with or without marks, untick **Penalise scripts and accents**. Punctuation is not a script: in “take x , then” the comma sits where a subscript would, but it is recognised by its shape and x scores as if it stood alone.

3 Looking a symbol up in the index

Many books have an index of notation that says where each symbol is introduced. **Look up in index** (or `Ctrl+Enter`) uses it:

1. It finds the index through the PDF’s bookmarks, or by its heading: a list of notation or symbols near the start of the book (for example right after the contents), an index on one of the last pages, or both.
2. It searches the index for the symbol and reads the page number printed next to it.
3. It jumps to that page and searches from there.

The printed page number is usually not the PDF page number, because front matter comes first. In this document, printed page 1 is PDF page 3. The panel shows where the index pointed, for example “Index → p. 3”. An “ $\approx$ ” after a page means the PDF page could not be confirmed and is an estimate.

This document has an index of notation on page 5. Try it: type `\mathfrak{F}` and press Ctrl+Enter.

4 Practice text

The following text is made up. It uses each symbol in the index of notation several times, in the ways they appear in real papers.

A toy filtration

Definition 1. A *toy filtration* is a family $\mathfrak{F} = (\mathfrak{F}_n)_{n \geq 0}$ of finite sets with $\mathfrak{F}_n \subseteq \mathfrak{F}_{n+1}$ for every n . We write $\mathcal{F}$ for the union of all $\mathfrak{F}_n$.

Definition 2. For a toy filtration $\mathfrak{F}$, let $\mathcal{F}$ be the collection of subsets of $\mathcal{F}$ that meet only finitely many of the sets $\mathfrak{F}_{n+1} \setminus \mathfrak{F}_n$.

Definition 3. A *stopping index* is a sequence $\tau_1 \leq \tau_2 \leq \dots$ of natural numbers. We write τ_j for its j -th term and $\mathcal{F}_{\tau_{j+1}}$ for the set $\mathfrak{F}_{\tau_{j+1}}$ seen as a subset of $\mathcal{F}$.

Lookalikes

The four letters $\mathfrak{F}$, $\mathcal{F}$, $\mathcal{F}$ and F are different symbols, and a search for one should not report the others. The same goes for $\mathbb{R}$ and R .

Spaces and classes

Definition 4. We write $\mathbb{R}$ for the real line and $\mathbb{R}^d$ for d -dimensional space. A map $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is *toy-smooth* if it is constant on each set of $\mathcal{F}$.

Definition 5. Let $\text{ID}(\mathbb{R}^d)$ be the class of toy-smooth maps, and $\text{ID}_2(\mathbb{R}^d)$ those that also vanish outside a bounded set.

Remark. Small occurrences are harder to find: in $\mathcal{F}_{\tau_j}$ the τ is in script size, and in $e^{\mathfrak{F}}$ the $\mathfrak{F}$ is. Lower **Min score** to about 0.85 to see them.

Exercises

Search for each of the following. The first page each one appears on is given, so you can check that the search order and the index lookup take you there.

type	first defined	notes
<code>\mathfrak{F}</code>	p. 3	also in Section 2
<code>\mathscr{F}</code>	p. 3	should not match $\mathfrak{F}$ or $\mathcal{F}$
<code>\mathcal{F}</code>	p. 3	
<code>\mathbb{R}^d</code>	p. 4	a bare $\mathbb{R}$ ranks lower
<code>\mathrm{ID}</code>	p. 4	also matches “ID” in IDEA, see Limitations
<code>\tau_j</code>	p. 3	also in script size

5 Limitations

- Scanned books work for the visual search. Their page numbers cannot be read for **Look up in index**, because a scan has no text layer.
- Content printed sideways (a table turned on its side) is found when the PDF has a text layer: the search then also tries the symbol turned the same way. In a scanned book it is not found. Rotating the view in the viewer is fine.
- Very small symbols (below about 6 pixels high at the search resolution) are skipped: at that size all shapes look alike.
- Everything happens in your browser: no page or query leaves your computer.
- A symbol that looks the same as ordinary letters is found in running text too: ID also matches the “ID” in IDEA, because the search compares shapes only.
- Commas and semicolons are told apart from subscripts by their shape, tuned on the font of this guide. In another font, a comma right after a symbol, as in “ x, y ”, may still be taken for a subscript and push the symbol below **Min score**. If a symbol you can see is not found, untick **Penalise scripts and accents**.

Index of Notation

$\mathfrak{F}$ toy filtration, 3
 $\mathcal{F}$ union of a filtration, 3
 $\mathcal{F}$ finite-meeting sets, 3
 τ_j stopping index, 3

$\mathbb{R}$ real line, 4
 $\mathbb{R}^d$ d -dimensional space, 4
 ID toy-smooth maps, 4
 ID_2 bounded toy-smooth maps, 4